

Application of adaptive neuro-fuzzy inference system in modelling home-based trip generation

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ABSTRACT

This study aims to investigate the feasibility of using the Adaptive Neuro-Fuzzy Inference System (ANFIS) and Multiple Linear Regression (MLR) for modelling home-based trip generation in Salfit City, Palestine. The research compares the performance of these two methods and provides insights into their efficiency for different trip purposes. The methodology involves developing separate trip generation models for various purposes, including total daily household trips generated (HBALL), home-based work (HBW), home-based education (HBE), and other home-based trips (HBO). The underlying principles and implementation details of the ANFIS architecture are also discussed. Objective evaluation metrics like Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R-squared are used to assess the effectiveness of each model. The results indicate that ANFIS performs well for modeling HBALL and HBO trips, which exhibit more complex behavior with wider data ranges and higher average daily trip counts. Compared to the MLR approach, ANFIS shows improved accuracy and closer predictions. For instance, when modelling HBALL trips, ANFIS achieved an RMSE of 1.4880, while MLR resulted in 1.7112, representing a reduction of 13.04%. However, for HBW and HBE trip purposes, where behavior is less complicated, MLR appears to be sufficient. The R-squared values obtained with MLR are sufficiently high to capture most of the trip variations, and the results between the two approaches are closely comparable. The R-squared value for MLR in HBE trips was almost identical to that of ANFIS, which was 96.65%. Similarly, for HBW trips, MLR achieved an R-squared value of 90.36%, while ANFIS performed slightly better with an R-squared value of 92.74%. In conclusion, ANFIS shows promise for modelling systems with complex behavior, while MLR remains a suitable option for less complicated scenarios. The study emphasizes the importance of exploring different modelling techniques in transportation research to identify the most appropriate approach for specific cases.

1. Introduction

The four-step travel demand forecasting process has been the paradigm used for modelling travel behaviour for decades. This process is still prevalent in many developing countries, including Palestine. The limited financial resources and the lack of reliable data are among the main reasons for not adopting more advanced approaches, such as tour- and activity-based models. The conventional four-step modelling process aims to forecast, within a rational framework, the current and future demand for travel by quantifying the relationships between urban activities and travel. This process involves four sequential steps: trip generation, trip distribution, mode choice, and trip assignment [1]. These steps are executed in sequence to convert urban activities into estimated number of trips.

The trip, in this context, is defined as a one-way journey made by an individual between two points, using a specified mode, or combined modes, of travel and for a defined purpose [2]. Trip generation is often considered as the first step in the four-step modelling process. Home-based trip generation specifically involves estimating the daily number of trips generated by households for various activities such as work, education, and other purposes [3]. This is achieved by developing mathematical models that formulate the relationships between household socioeconomic characteristics, such as size, employment status, and income level, and the corresponding number of trips generated [4]. It is crucial for these mathematical approaches to be accurate and reliable in modelling trip generation, as it forms the basis for the subsequent steps of travel demand modelling.

The traditional Multiple Linear Regression (MLR) model is one of the

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most commonly used approaches in modelling trip generation due to its simplicity and generally acceptable performance [5]. MLR models generally assess the correlation and the strength of association between the generated trips and a set of explanatory variables that determine the trip-making behaviour, such as household socioeconomic characteristics [3].

The linear structure of MLR and the correlation effects between the explanatory variables may create estimation problems and yield unrealistic models [6]. This is quite expected when the underlying relationships among these variables are complex, nonlinear, or not fully understood, such as the case of the trip-making behaviour. Alternative approaches, such as the category analysis technique, have been used to avoid the assumption of linear, additive relationship between trip generation and its explanatory variables. However, this approach has no statistical justification where only few and specific variables could be considered [7].

It is important to note that the trip-making behavior involves complex decision-making processes characterized by uncertainty and ambiguity, making it difficult for conventional techniques to accurately capture these uncertainties without significant deterioration in the model performance [8]. Alternatively, the use of advanced machine learning algorithms, such as artificial neural networks and decision trees, has attracted much attention recently. These techniques, which combine elements of self-learning, adaptation, and self-organization, enable effective elaboration for modelling complex and nonlinear behaviour [9].

The Adaptive Neuro-Fuzzy Inference System (ANFIS) is an advanced machine learning approach that integrates both the Artificial Neural Network (ANN) and the Fuzzy Inference System (FIS) into a single framework [10]. This approach has been proven to be superior to traditional techniques, as it can handle nonlinear and complex relationships between inputs and outputs, and model the uncertain and imprecise information in the input–output data, which are common in many real-world applications [11]. Despite its potential, ANFIS has seen limited application in travel demand in general, and trip generation at specific, as highlighted in the literature review section. There is still a shortage in previous studies that focus on using ANFIS for trip generation modelling. This dearth of research makes it challenging to draw a clear conclusion about the performance of the ANFIS approach in comparison to traditional methods regarding trip generation modelling.

This paper aims to evaluate the potential of using the ANFIS in modelling home-based trip generation, and compare its performance relative to the conventional MLR approach. The underlying principles of the ANFIS architecture and its implementation methodology are comprehensively discussed. The results of the two approaches considering the same dataset are compared using a number of statistical measures such as the Root Mean Squared Error (RMSE), the Mean Absolute Error (MAE), and coefficient of determination (R-squared).

The application and comparison have been explored through the development of four main types of disaggregated home-based trip generation models, that vary in the level of complexity, for the Palestinian city of Salfit. These models are 1) the HBALL model to estimate the total number of daily trips generated by a household; 2) the HBW model to estimate daily Home-Based Work trips; 3) the HBE model to estimate daily Home-Based Education trips; and finally, 4) the HBO model to estimate the number of daily Home-Based Other trips, which are generated not for work or education purposes. However, non-home-based trip generation models are outside the scope of this study.

The merits of ANFIS approach and its limited application in travel demand have encouraged conducting this study. Moreover, due to the fuzzy nature of many datasets, such as the case of travel demand, it may be more appropriate to construct and evaluate prediction models using FIS [12]. A more realistic comparison could be achieved by considering also other advanced machine learning approaches, such as the artificial neural network itself. However, this is not the case here, as this study does not aim primarily to find the best modelling approach for trip

generation, but aims to explore the performance of ANFIS for this task. This study addresses the development process of the indicated trip generation models, and chooses the MLR as a benchmark to assess the efficacy of the proposed ANFIS models.

It is to be indicated that there is a scarcity of studies that thoroughly evaluate the performance of ANFIS by creating multiple trip generation models for different trip purposes with the same dataset and application context, which motivates the need for this study. This is different from the limited published studies mentioned in the literature review section, where only a single type of trip generation model was developed using ANFIS and compared with MLR.

The remainder of this paper is organized as illustrated hereafter. Section 2 provides a brief overview of the ANFIS architecture and reviews related works and literatures. Section 3 outlines the methodology used to develop the intended models considering both MLR and ANFIS. Section 4 presents the study area, application context, and data collection and description. In Section 5, the associated results and the models developed using both approaches are presented. Finally, Section 6 concludes the study and provides recommendations for future work.

2. Literature review

Several statistical approaches have been adopted for modelling trip generation, such as the multiple linear regression, logistic regression, negative-binomial models and ordered probit models. The application of these models in trip generation, their benefits and limitations, have been evaluated and compared through several relevant studies such as those of Lim & Srinivasan [13] and Chang et al. [5]. In general, the MLR was found to be the most widely used approach for this purpose due to its simplicity and acceptable performance. Moreover, a comprehensive review of the different approaches used for trip generation modelling over the past two decades can be found in Mukherjee & Kadali [14].

Many researchers in various scientific disciplines have demonstrated the effectiveness of ANFIS approach in modelling systems with complex and nonlinear behaviour, as well as its superiority over traditional techniques. In this regard, this section gives a brief overview of the ANFIS architecture and presents its applications, along with those of ANN and FIS, in transportation research and trip generation at specific. This section also summarizes recent studies conducted in Palestine concerning trip generation.

2.1. Adaptive Neuro-Fuzzy Inference System

ANFIS was first proposed and developed by Jang in 1993. It is an advanced modelling technique that blends fuzzy logic with neuro-computing and genetic algorithms in order to model complex nonlinear systems. ANFIS can construct an input–output model using stipulated input–output data and human knowledge in the form of fuzzy if-then rules [15]. It is basically a FIS implemented within the framework of ANNs.

The ANN is a machine learning model that simulates the way the human brain analyzes and processes information. Using specific learning algorithms, ANN learns to perform tasks by considering input–output dataset without being explicitly programmed with task-specific rules. The ANN can adapt itself to organize its structure when presenting training examples [10]. The ANN consists of multiple interconnected artificial neurons that are organized systematically to form a layered pattern. In short, the architecture of ANN usually consists of three different layers: input, output, and one or more hidden layers. Each layer contains several processing neurons, each with a specific function that may have fixed or adjustable parameters [10].

The FIS, on the other hand, is a popular computing framework that simulates the human way of thinking and reasoning. The fuzzy logic is based on the degrees of truth, which represent the degree of membership in a fuzzy set, rather than the usual true or false (i.e., 0 or 1) in the classical set. In the fuzzy logic, there can be many truths between 1

(completely true) and 0 (completely false). The fuzzy set was first proposed by Zedeh in 1965 [16]. It has been proved to be an efficient way of dealing with uncertain or vague information and for defining linguistic variables such as low or high.

In a FIS, knowledge about the problem domain is encoded in the form of if-then rules, where the antecedent part of the rule describes the input conditions, and the consequent part defines the output action [17]. Membership Functions (MFs) are used to assign a membership value to each input variable. For each input or output variable, multiple MFs can be identified, each reflecting a particular fuzzy set that represents a particular linguistic term. MFs can have different shapes, such as triangle, trapezoidal, or Gaussian [18].

Generally, the FIS converts the crisp inputs into fuzzy inputs with values ranging from 0 to 1 through the fuzzification interface based on the MFs. The fuzzy values are then processed using the rule bases, with the resulting fuzzy output value being de-fuzzified to a continuous real output value [18]. The Takagi-Sugeno FIS is one of the most commonly used and well-known FIS models in the engineering applications of ANFIS [19].

Accordingly, ANFIS is multilayer feed-forward artificial neural network that is functionally equivalent to the Takagi-Sugeno FIS model. It is an integrated system that combines both the learning capabilities of ANN and the reasoning capabilities of FIS into a single framework [17]. ANFIS initiates a FIS using an input-output training dataset and then uses the learning capability of ANN to optimize the fuzzy rules and parameters of the MFs. The optimization process is accomplished through repeated training with a designated learning algorithm over multiple epochs. Training epochs number refers to the number of times the system has undergone training or the iterations of the training process.

The learning algorithm attempts to reduce the difference between the actual and predicted values by typically using the RMSE as a measure of error. In this study, a hybrid learning algorithm has been considered. This algorithm combines a back-propagation gradient descent method for the input MF parameters, and a least squares estimation approach for optimizing the output MF parameters. Further information on this is found in literature (e.g., [15;20]).

To simplify the concept, assume that a typical first-order Takagi-Sugeno fuzzy inference model, as shown in Fig. 1(a), represents a home-based trip generation model with two input variables. The first is the household size (x), and the second is the income level (y) for instance. Each variable has two Gaussian MF, one for high values and one for low values, and two output linear MFs (f). The rule set of this model can be expressed using two basic fuzzy if-then rules:

Rule 1 : If x is A_1 and y is B_1 , then $f_1 = p_1x + q_1y + r_1$

Rule 2 : If x is A_2 and y is B_2 , then $f_2 = p_2x + q_2y + r_2$

Where A_1, A_2, B_1 and B_2 are the MFs of the inputs (x, y), respectively, and (p_i, q_i, r_i) for ($i = 1, 2$) are the linear parameters of the output MFs [15]. The corresponding typical ANFIS architecture is shown in Fig. 1

(b). It consists of five layers ANN as the following:

Layer-1 represents the fuzzification step, which contains the input MFs. The neurons in this layer are adaptive with adjustable parameters. The output of this layer is the membership value of each fuzzy set between 0 and 1, which is given by:

$$O_i^1 = \mu_{A_i}(x), \text{ for } i = 1, 2, \text{ or } O_i^1 = \mu_{B_i}(y), \text{ for } i = 3, 4 \quad (1)$$

where x (or y) is the crisp input to node i (e.g., income level), A_i (or B_i) is the associated fuzzy set (e.g., the 'large' set of the input), and μ_{A_i} (or μ_{B_i}) is the MF of the fuzzy set A_i (or B_i). The Gaussian MF in this example can be defined using the standard deviation (σ) and the mean value (μ), changing these parameters will change the shape of the MFs.

Layer-2 called the rule layer, which simply performs a multiplication task between the output of the first layer. Each node output in this layer represents the firing strength of each rule, and can be represented as:

$$W_i = \mu_{A_i}(x) \times \mu_{B_i}(y), \text{ for } i = 1, 2 \quad (2)$$

Layer-3 performs a normalization task among the output of the second layer. The outputs of this layer, which is called normalized firing strengths, are given by:

$$\bar{W}_i = W_i / (W_1 + W_2), \text{ for } i = 1, 2 \quad (3)$$

Layer-4 represents the defuzzification step and contains the output MFs. The neurons in this layer are adaptive with adjustable parameters. The outputs of this layer can be expressed as:

$$O_i^4 = \bar{W}_i f_i = \bar{W}_i(p_i x + q_i y + r_i), \text{ for } i = 1, 2 \quad (4)$$

Layer-5 which computes the overall output (i.e., the generated trip number) by performing a simple summation task. The overall output is given as:

$$\text{Overall Output} = f = O_i^5 = \sum_i \bar{W}_i f_i = \frac{\sum_i W_i f_i}{\sum_i W_i}, \text{ for } i = 1, 2 \quad (5)$$

2.2. ANN use in transportation and trip generation

The effectiveness of the application of ANN in the transportation sector and travel demand has been demonstrated through various studies. For example, Karlaftis & Vlahogianni [21] and Pamula [9] compared the use of statistical methods and neural networks in transportation research. They found that neural networks are more flexible for modelling complex datasets with nonlinearities or missing data. Recently, a lot of studies have been also conducted in this field such as Wang et al. [22], Koushik et al. [23], Yan et al. [24], and Pamula & Zochowska [25], who explored the applications of machine learning in transportation systems, activity-travel behaviour, direct demand modelling of ride-sourcing services, and estimating the OD Matrix using deep learning, respectively.

In trip generation, Tillema et al. [26], for example, compared the performance of ANN and the commonly used MLR in modelling trip

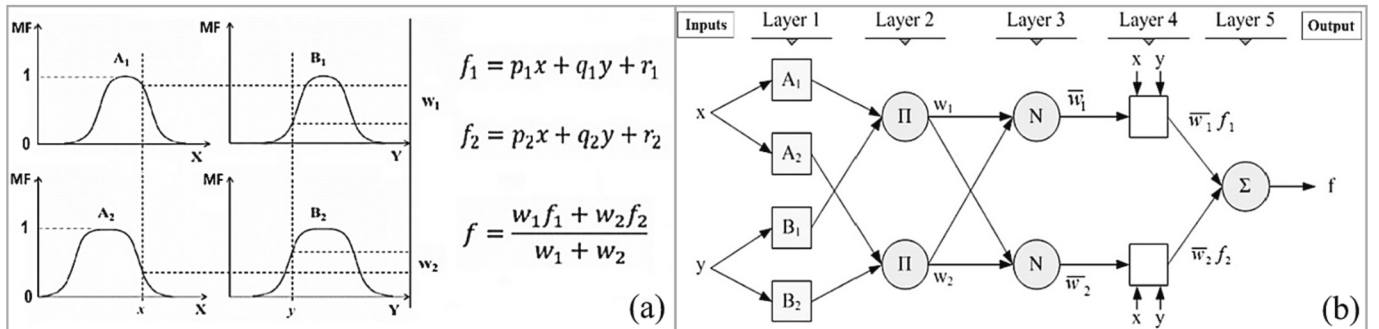


Fig. 1. (a) Takagi-Sugeno Fuzzy Inference Model; (b) The Equivalent ANFIS Architecture. [15].

generation using a synthetic dataset generated from Dutch national travel diary data. The comparison was made using the RMSE metric, and they found that ANN can be successfully used for modelling trip generation and usually able to outperform traditional methods. Goel & Sinha [27] also showed the effectiveness of ANN in modelling trip generation in Meerut City, India, as it was able to handle the nonlinear relationship between socioeconomic and transportation variables.

Arliansyah and Hartono [28] used a Radial Basis Function Neural Network (RBFNN) to model trip generation in Palembang, Indonesia, and compared it with the regression approach. The RBFNN performed better, with a coefficient of determination (R-squared) of 74.7%, compared to the regression model, which had an R-squared value of 45.9%. Etu and Oyedepo [29] also developed a trip generation model for Akure, Nigeria, using both RBFNN and MLR. The study found that the RBFNN produced more accurate trip generation forecasts compared to the MLR. Kabakus & Tortum [30] provided comparative analysis of trip generation models according to household characteristics for a number of provinces in Turkey.

Several recent studies in this field have been also conducted such as Naser et al. [31] who analyzed the performance of an ANN model for Nasiriyah, Iraq and compared it with the same model developed using MLR. The ANN model showed better results with an R-squared of 94.8% compared to 87.1% for MLR. Li et al. [32] used gradient boosting algorithm for urban trip generation forecasting. Xiao et al. [33] used the convolutional neural network for trip generation prediction. Moreover, Lafta & Ismael [34] developed a trip generation model for Al-Karada region in Baghdad City, Iraq, incorporating socioeconomic features of 1,050 households. The ANN prediction model was found to be more accurate, with an average accuracy of 83.72%, compared to 72.46% for MLR.

2.3. FIS and ANFIS use in transportation and trip generation

The use of fuzzy logic in trip generation was introduced by Kalic and Teodorovic [35]. The number of generated trips was determined using FIS, the artificial neural network, and MLR. In this study, the fuzzy logic was able to provide the closest estimate of the actual number of trips generated in the given area. Teodorović [36] further explored the application of FIS in solving various transportation problems including trip generation, trip distribution, mode choice, and trip assignment. The author found that the fuzzy logic, and the FIS in general, represent a promising approach for modelling transportation problems characterized by uncertainty and imprecision.

In trip generation, Rassafi et al. [37], for example, developed a FIS for predicting the rate of trips generated in Mashhad, Iran. Simha [38] also applied a fuzzy rule-based model for trip generation prediction in the urban and suburban areas of Guntur City, India. Both studies used household socioeconomic factors to develop trip generation models and compared the results to those obtained from MLR. The authors addressed the limitation of the MLR. They found that the FIS was more suitable for trip generation estimation compared with the MLR.

The study by Stojčić [11] conducted a comprehensive review of ANFIS applications from 1993 to 2018 across various transportation fields, including modelling, safety, traffic control, and congestion. The author found that ANFIS typically demonstrates improved performance compared to conventional methods for modelling non-linear systems. In addition, several studies have addressed the application of ANFIS in modelling mode choice, such as Andrade et al. [39], Tortum et al. [40], and S et al. [41]. In these studies, ANFIS was also able to outperform traditional techniques in modelling the nonlinear behavior of mode choice. Wang et al. [42] showed through empirical results that the ANFIS model is better at forecasting automobile sales in Taiwan than traditional techniques.

However, despite the superiority of the ANFIS approach over traditional methods, its application in trip generation modelling has been limited to particular few studies such as Příbyl & Goulias [43]. In this

study, the authors developed an ANFIS model to estimate the number of trips generated per person per day using data collected from South Perth, Australia. The authors compared their results with two traditional methods - linear regression and negative binomial models. The ANFIS was able to outperform these two traditional techniques, especially when using Gaussian type membership functions and training the model using the backpropagation learning algorithm for 100 epochs.

Ahmadpour et al. [44] utilized ANFIS to model work trip production in Adelaide, Australia, based on household characteristics. They found that ANFIS was beneficial as it incorporated both human knowledge in the form of linguistic terms and input-output data in the modelling process. Seyedabrishami & Shafahi [45] developed an expert-guided ANFIS to tackle uncertainties in trip generation estimation for Shiraz, Iran. The model outperformed traditional approaches in terms of accuracy and generalization capabilities. Moreover, Mahdavi & Mamdoohi [46] developed an ANFIS for modelling trip generation based on land use data and socioeconomic characteristics for Qazvin City, Iran. A comparison was performed with linear regression approach, and the result showed that the ANFIS has superiority over the linear regression.

In the study of Akinfala et al. [47], the effectiveness of MLR, ANN, and ANFIS for trip generation modelling was evaluated. A daily household trip generation model was developed based on household socioeconomic characteristics for the city of Lagos, Nigeria, using these three approaches. The performance of each model was evaluated using several statistical metrics such as Mean Squared Error (MSE) and R-squared. The MSE values for regression, ANN, and ANFIS were 1.15, 1.16, and 1.15, respectively, while the R-squared value was 71% for all approaches. In contrast, the authors concluded that ANN and ANFIS are not superior alternatives to MLR for trip generation modelling and that the performance gains from these models are minimal and can be disregarded.

Despite that the above literature expanded over a period of more than 20 years, there is still a shortage in previous studies that focus on using ANFIS for trip generation modelling. This dearth of research makes it challenging to draw a clear conclusion about the performance of the ANFIS approach in comparison to traditional methods. Moreover, in each of the above studies, only a single type of trip generation model was developed using ANFIS and compared with MLR. However, there is a scarcity of studies that thoroughly evaluate the performance of ANFIS by creating multiple trip-generation models for different trip purposes with the same dataset and application context, which has motivated the need for this study.

2.4. Trip generation modelling in Palestine

In Palestine, few studies have been conducted for modelling trip generation. However, the most relevant study is that of Dodeen [48]. The author successfully developed several home-based trip generation models for Jericho City. These models were estimated based on the social and economic characteristics of households in the area, and developed using the linear regression approach. Another related study in this context is the study of Amer [49]. Using the trip generation models developed by Dodeen [48], the author investigated the ability for spatial transferability of these models from Jericho City to the target area of Salfit City.

However, the study of Abu-Eisheh & Irshaid [50] is the first and only one in Palestine that applied the ANFIS approach for trip generation modelling. The authors developed a trip generation model that estimates the total number of daily trips generated by a household for Salfit City, and evaluated their results as relative to the linear regression approach. ANFIS was able to outperform the regression technique and produce a more accurate model. The authors recommend using Gaussian membership functions for input variables, and utilizing a hybrid learning algorithm for optimizing the system. However, in this study the issue of model overfitting was not addressed. The authors made an arbitrary choice for the number of training epochs without monitoring the testing error, which could raise concern about the reliability of the results.

Model overfitting is a common issue in machine learning applications which will be further discussed.

This study provides further investigation of the potential of using ANFIS in modelling home-based trip generation with rectified and clear methodology, and evaluates its performance as compared with the MLR approach. The comparison has been conducted additionally between three main types of trip generation models, besides modelling the overall number of generated trips, with different levels of complexity, considering the same database and application context.

3. Research methodology

The aforementioned four trip generation models have been developed here using both MLR and ANFIS. The methodology for each approach used to develop each model is explained below. The performance evaluation and comparison for each model type, as well as the validation procedures, are also discussed.

3.1. Development of models using MLR approach

In this step, each model has been estimated and calibrated using the MLR approach. The backward stepwise regression method is employed to identify the most significant explanatory variables for the model. Initially, all available socioeconomic variables are included, and then, iteratively, the variable with the least impact on model fit is removed until no further deletions can occur without a statistically significant loss of fit. Table 1 in Section 4 presents the selected variables used in this study. The estimation of coefficients for the chosen explanatory

Table 1
Models Explanatory and Dependent Variables Description.

Variable	Description	Average	St. Deviation	Min-Max
Explanatory Variables (Models Inputs)				
SIZE	Number of persons (household size)	3.99	1.749	1–9
M	Number of males in the household	2.01	1.181	0–6
F	Number of females in the household	1.98	1.096	1–6
EMP	Number of persons who are employed	1.38	0.899	0–4
EDU	Number of persons who are receiving education	1.63	1.529	0–7
AGE _A	Number of persons aged under 16 years	1.24	1.373	0–6
AGE _B	Number of persons aged between 17 and 30 years	1.14	0.977	0–4
AGE _C	Number of persons aged between 31 and 50 years	0.92	0.805	0–2
AGE _D	Number of persons aged between 51 and 64 years	0.51	0.728	0–3
AGE _E	Number of persons aged above 65 years	0.19	0.485	0–2
DRV	Number of licensed drivers	1.18	0.964	0–6
CAR	Number of cars owned by a household	0.54	0.605	0–3
INC	Monthly household Income (Thousand New Israeli Shekel)	4.74	2.993	0.5–30.0
HHTYP	Household Type: 1 if independent residence, 0 if apartment	83.8% Independent Residence		
Dependent Variables (Models Output)				
HBALL	Number of total daily trips generated by household	6.41	3.05	2–18
HBW	Number of daily work trips generated by household	1.56	1.044	0–5
HBE	Number of daily education trips generated by household	1.59	1.546	0–7
HBO	Number of daily other trips generated by household	3.26	1.774	0–11

variables was performed using the ordinary least-squares method. This approach evaluates the significance of each explanatory variable and its associated coefficient through various commonly used statistical tests, which also assess the overall fitness of the regression model.

This study performs relevant statistical tests to assess the validity of the developed models. To address the potential issue of multicollinearity among the numerical variables, Pearson's correlation and Variance Inflation Factor (VIF) are used. The significance of each individual coefficient is tested using the well-known *t*-test. The *F*-test is used to evaluate the overall significance of the model. Analysis of variance (ANOVA) is performed to measure the explanatory power of the variables. Finally, the goodness of fit is evaluated using the RMSE, MAE and R-squared.

3.2. Development of models using ANFIS approach

In this step, an initial Takagi-Sugeno FIS is generated for each model using the same training input–output dataset and the same input variables as in the previous step. The same input variables have been considered for this step to ensure a fair comparison with the MLR approach, to avoid the linear correlation effects which might be exist among some variables, and to reduce the computational cost of the desired models.

Gaussian type MFs, which defined using standard deviation (σ) and mean value (μ), have been used for all input variables in this study as recommended by Příbyl & Goulias [43] and Abu-Eisheh & Irshaid [50]. In this step, it is necessary to determine the appropriate number of MFs that should be used for each input variable. The selected number of MFs will be used to ensure proper partitioning of the input range of each variable, and this will determine the MF parameters. In this study, three Gaussian MFs are selected for each input variable to balance between accuracy and cost, while only two functions are used for input variables with only two values such as the household type which could be 1 for independent residence and 0 if apartment.

For each combination of input variable MFs, a fuzzy rule and a corresponding linear output MF are then created to generate the expected decision rules. For instance, if two input variables (e.g., household size and income) each have three MFs (e.g., small, medium, and large income MFs), 9 fuzzy rules ($3 \text{ MFs} \times 3 \text{ MFs} = 9 \text{ MFs} \times 3 \text{ MFs} \times 3 \text{ MFs} = 27$) and 9 linear output MFs will be created. The parameters of each linear output MF are initialized to zero prior to the optimization process. The FIS is then optimized using a hybrid learning algorithm, which adjusts the parameters of the input and output MFs using the training dataset, as outlined in Section 2. The algorithm trains the FIS multiple times (epochs) until it reaches its optimal structure.

The number of training epochs selected is crucial to the success of the developed ANFIS model. An excessive number of training epochs can lead to overfitting the training data, hindering the generalization ability of the model. The goal of the training algorithm is to minimize the difference between the actual data and the predicted output during each epoch. Over time, the training and testing errors will decline, but eventually, the model may start overfitting and capturing the randomness in the training dataset. At that point, the training error may continue to decrease or remain stable, while the testing error will start to rise. However, this issue was not considered in the study of Abu-Eisheh & Irshaid [50] as mentioned earlier.

To avoid the overfitting problem, it is common practice to divide the collected dataset into 80% for training and 20% for testing purposes [51]. The optimal number of training epochs is the one that results in the lowest possible training and testing error. In this study, nearly 80% of the collected dataset is designated for training the model and is used to calculate the training error after each iteration (epoch). The remaining 20% of the dataset that was not used for training is used to determine the testing error. The training and testing datasets are thoroughly discussed and referenced in Section 4. The flowchart in Fig. 2 summarizes the steps taken to develop and train the ANFIS model as outlined above.

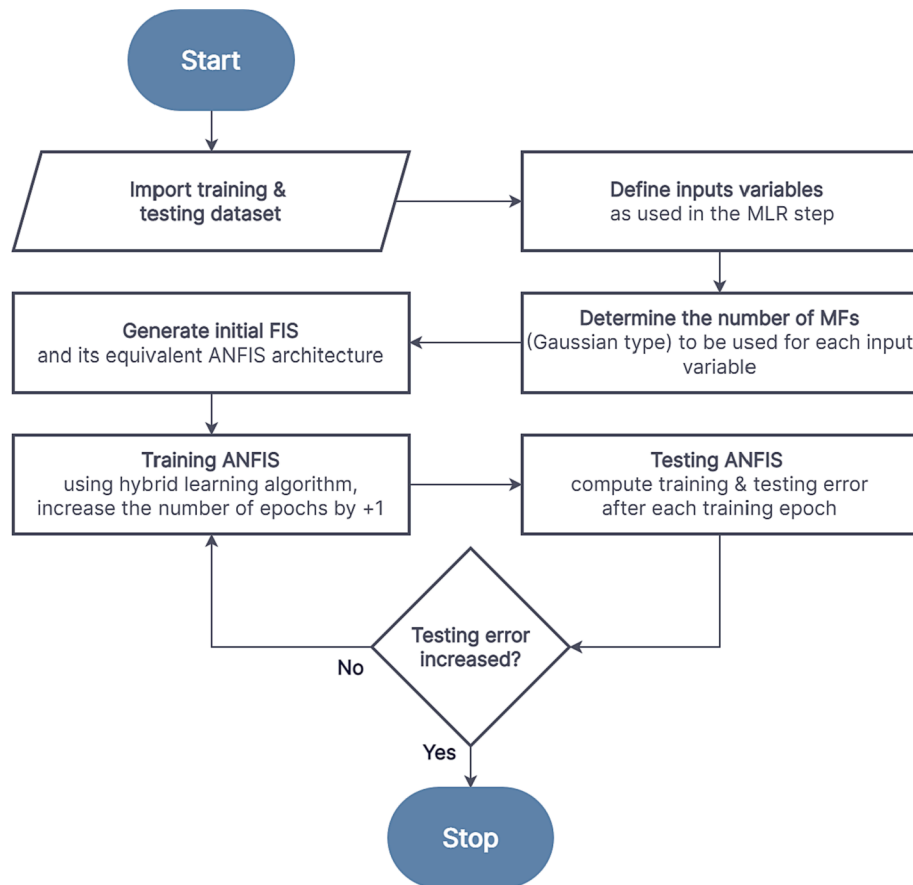


Fig. 2. Flowchart for Developing and Training an ANFIS for Each Model Type.

3.3. Performance comparison and validation

The predictive accuracy of the models in this study was thoroughly evaluated based on three widely adopted statistical measures, RMSE, MAE, and R-squared. These measures provide a comprehensive way to compare the performance of each approach and offer valuable insight into the quality of the predictions made by the models. As known, low RMSE and MAE values indicate a high degree of accuracy and a close match between the actual and predicted values, while a high R-squared value suggests a strong linear relationship between the independent and dependent variables, and therefore a good fit of the model to the data. These measures have been widely used in machine learning applications and have been recommended by various studies as well as the developer of the ANFIS approach, Jang [15].

A separate testing dataset was also used for validation purposes. This dataset was not involved in the estimation of the model of interest. The validation process, which is a crucial part of model development, involves a comparison between the actual values and the predicted values from the estimated models, to ensure that they met acceptable standards of conformity. The use of a separate testing dataset for validation purposes further strengthens the robustness of the results obtained from this study. The selected testing dataset is thoroughly discussed and referenced in Section 4. In conclusion, the approach with the best performance is determined by lower RMSE and MAE values, higher R-squared values, and predictions that were closest to the actual values.

4. Study area and data

Salfit City is considered as the context of application for this study. It is a small Palestinian community with a total population of 10,911 persons, distributed among a total number of 2,527 households. This is

according to the census conducted in 2017 by the Palestinian Central Bureau of Statistics (PCBS). Considering the resource limitations of data collection in Palestine, Salfit City was selected due to the availability of the dataset needed to perform this study. This dataset has been thoroughly collected in the study of Amer [49], in which the author conducted a household-based trip generation study for the city, as mentioned earlier.

In the study of Amer [49], the required dataset has been collected by dividing the study area into six internal homogeneous traffic analysis zones and conducting personal face-to-face interviews with randomly selected households. The zones were used to ensure an even distribution of the samples across the study area. A total of 256 households were interviewed and used to estimate and train the desired models. Moreover, an additional 53 households were also interviewed and used for validation and testing purposes.

It is essential to highlight that while the sample size may appear small, it was determined to be the minimum required to cover 10% of the households in the city, ensuring its representativeness. However, the justification for using ANFIS under these circumstances lies in its capacity to effectively handle complex behavior and wider data ranges, even with limited datasets. By employing ANFIS, the aim is to capture the underlying relationships within the data and provide meaningful insights despite the constraints posed by the sample size.

Each input data vector in the collected sample represents one household and contains data regarding its socioeconomic characteristics and the associated number of trips generated. The socioeconomic characteristics of the household are represented by relevant explanatory variables, which served as inputs to the model. The number of trips generated is also represented by four dependent variables that constitute the model outputs, as described in Table 1.

The input variables in this study are consistent with those used in the

study of Abu-Eisheh & Irshaid [50]. The difference here is the inclusion of three additional dependent variables, as mentioned earlier. It is worth mentioning that the majority of the study area, over 80%, is comprised of residential areas and therefore, variables related to land use were not considered in the analysis. Moreover, the study area that is characterized as a medium sized city, has population with similar cultural and traditional backgrounds, therefore, it is safely assumed that households generate trips in the same pattern regardless of their location.

5. Models development and comparison

As shown in Table 1, the HBALL model estimates the number of all possible household trips generated during a typical workday. This can range from a minimum of 2 trips to a maximum of 18 trips per day. The HBW, HBE, and HBO models, on the other hand, estimate the number of daily trips generated for work, education, and other trip purposes, respectively. These estimates range from a minimum of 0 trips to a maximum of 5, 7, and 11 trips per day, respectively. The following subsections present the resulted models using both MLR and ANFIS.

5.1. Resulted models using MLR

Equations (6), (7), (8), and (9) suggest the best explanatory variables and corresponding coefficients for estimating the number of daily trips generated for HBALL, HBW, HBE, and HBO, respectively. These equations represent the best estimates for the desired models resulting from the regression analysis previously presented in Section 3.1. Table 2 displays the regression results, including the RMSE, MAE, and R-squared values for each model.

$$\text{HBALL} = 2.388 + 1.046 \text{EMP} + 1.237 \text{EDU} + 0.424 \text{DRV} \quad (6)$$

$$\text{HBW} = 0.875 \text{EMP} + 0.153 \text{AGE}_B + 0.280 \text{CAR} \quad (7)$$

$$\text{HBE} = 0.984 \text{EDU} \quad (8)$$

$$\text{HBO} = 0.415 \text{SIZE} + 0.289 \text{DRV} + 1.218 \text{HHTYP} \quad (9)$$

The HBALL model of Equation (6) is consistent with the one in Abu-Eisheh and Irshaid's [50] study, which was expected as the same procedure for regression modelling was followed in this study. In this model, the number of trips tends to increase with the increase in the number of employed persons in the household (EMP) with a coefficient of 1.046, persons going to school (EDU) with a coefficient of 1.237, and those with a driving license (DRV) with a coefficient of 0.424.

As Table 2 shows, the EDU variable has the highest correlation with the number of trips (i.e., highest t-value), followed by EMP, and with a lower correlation with DRV. This is intuitively true, as most daily household trips are likely to be made by those who are in education or employed. The intercept of the model has a high t-value compared with the other variables, making it a crucial component. The household is likely to make at least two trips per day even if all explanatory variables are set to zero.

Table 2
Regression Results for Each Model.

Model Type	Variables	Coef.	t-value	Sig.	VIF	F-value	R-Squared	RMSE	MAE
HBALL	Intercept	2.388	10.721	0.000	–	161.982	65.85%	1.7112	1.3882
	EMP	1.046	7.945	0.000	1.287				
	EDU	1.237	16.926	0.000	1.029				
	DRV	0.424	3.452	0.001	1.262				
HBW	EMP	0.875	21.605	0.000	1.312	790.181	90.36%	0.5932	0.4163
	AGE _B	0.153	3.813	0.000	1.228				
	CAR	0.280	4.534	0.000	1.086				
HBE	EDU	0.984	85.556	0.000	1.000	7319.795	96.63%	0.4035	0.1430
HBO	SIZE	0.415	8.307	0.000	4.847	351.584	80.65%	1.5778	1.2825
	DRV	0.289	2.809	0.005	2.559				
	HHTYP	1.218	5.497	0.000	4.362				

The HBW model, on the other hand, increase with the increase in the number of employed persons in the household (EMP) with a coefficient of 0.875, the number of young household members with ages between 17 and 30 years with a coefficient of 0.153, and the number of cars owned by the household with a coefficient of 0.280. As Table 2 indicates, the variable EMP has a strong correlation with HBW, which is expected as work trips are usually made by employed or young individuals. The other variables are also significant, but have a less impact on HBW trips compared to EMP. The intercept was not included in this model, as there would be no work trips if the explanatory variables in the HBW model were set to zero, as expected.

The HBE model is highly influenced by the number of household members who are receiving education with a coefficient 0.984, as shown in Table 2. When there are no members of the household pursuing education, there will be no education trips. HBE is correlated only with EDU with no intercept as expected.

Finally, the HBO trips tend to increase with the increase in the household size with a coefficient 0.415, the number of individuals with a driving license with a coefficient 0.289, and the presence of an independent household type with a coefficient 1.218. Table 2 shows that the HBO trips is strongly correlated with the size and type of the household. An increase in the values of these two variables leads to an increase in the number of household members, which in turn results in more trips.

Generally, the directly proportional and positive the dependent and explanatory variables in the four models discussed above is consistent and makes logical sense. The regression results in Table 2 indicate a good explanation of the data variability, with a reasonable R-squared value for each model. Additionally, the absence of multi-collinearity issues is confirmed by the Pearson's correlation and VIF values, which are below 0.50 and 10 respectively for each of the models.

The t-value for each coefficient in each model, as displayed in Table 2, exceeds the required threshold of 2 and the p-value for each coefficient in each model is lower than the desired significance level of 0.01, indicating that every chosen explanatory variable for each model is statistically significant. Furthermore, the F-test for the overall significance of each model demonstrates that the coefficients jointly contribute to the model. This rejects the null hypothesis for each model, which indicates that all regression coefficients are equal to zero, or in other words, that the model has no predictive capability.

5.2. Resulted models using ANFIS

The optimal ANFIS structure for the HBALL model, which generates the lowest possible training and testing RMSE, consists of three Gaussian-type MFs for each input variable (EMP, EDU and DRV), 27 fuzzy rules with 27 linear output MFs. This was obtained by training the model over 80 epochs. For this model, an initial Takagi–Sugeno FIS was generated using the same training dataset and input variables as in the regression step. Three Gaussian-type MFs for each input variable were then identified, each reflecting a fuzzy set with a specific linguistic term. For example, for the EMP input variable, low, medium, and large EMP

were represented by three MFs. Consequently, 27 fuzzy rules (i.e., $3 \times 3 \times 3 = 27$) were created with 27 linear MFs for the output.

The hybrid learning algorithm was then utilized to optimize the associated parameters of the input and output membership functions. The model has been trained by the learning algorithm for 80 epochs, before the model starts overfitting the data as discussed earlier. Table 3 shows the minimum training and testing RMSE that was achieved. This model is different from the one obtained by Abu-Eisheh & Irshaid [50], where their model was trained arbitrarily for over 1000 epochs without monitoring the training and testing errors simultaneously, which could lead to an overfitted model and raise questions about its reliability.

The ANFIS architecture for the HBALL model is shown in Fig. 3. This model is composed of five layers. The first layer consists of the input variables, the second layer comprises of the input MFs, the third layer encompasses the rule layer in the FIS and performs normalization, the fourth layer features the output linear MFs for each rule, and the fifth layer presents the final crisp HBALL value. Moreover, Fig. 4 shows the transformations of the input MFs for each input variable, from their initial to optimized forms, in the HBALL model.

The same procedures are repeated for the remaining models. For HBW, the optimal ANFIS structure consists of three Gaussian MFs for each input variable (i.e., EMP, AGE, and CAR), and was obtained by training the model for 20 epochs. On the other hand, the optimal structure for the HBE model was obtained using three Gaussian MFs for the EDU input variable, and achieved by training the model for 10 epochs. Finally, for HBO, the optimal configuration was achieved with three Gaussian MFs for SIZE and DRV, two Gaussian MFs for HHTYP, and 100 training epochs. Two Gaussian MFs were identified for HHTYP, as it could be either 1 if the household is independent residence or 0 if it is an apartment. The results associated with the optimal ANFIS configuration for each model, as long as the RMSE and MAE for training and testing samples, are presented in Table 3. The parameters of the input MFs before and after optimization are shown in Table 4.

The learning algorithms optimizes the parameters (f_i) associated with each linear output MF, as outlined in Section 2. These parameters are initially set to zero before the start of the training process, and the algorithm assigns specific values to them until the optimal ANFIS model is obtained. For example, in the case of HBALL model, each output MF has the form: $f_1 \text{ EMP} + f_2 \text{ EDU} + f_3 \text{ DRV} + f_4$. There are 27 output MFs, each associated with a particular rule, as shown in Fig. 3. Each output function has four parameters that need to be optimized. However, due to the large number of parameters associated with each model type, it is not feasible to present their values in this paper.

5.3. Performance comparison and validation

The comparison of the performance of each model developed using MLR and ANFIS is presented in Table 5. This comparison is based on the RMSE, MAE, Residual Sum of Squares (RSS), and R-squared measures. The testing sample with the additional 53 households is used also for validation purpose. The number of trips predicted by each model and the actual values for both modelling approaches are within acceptable conformity. Table 5 also summarizes a comparison of the RMSE, MAE, means, and standard deviations of actual daily trips and the predicted trips by each model type for the testing sample. The best approach is determined by lower RMSE, MAE, and RSS, higher R-squared, and

predictions that are closest to the actual values.

In this study, the complexity of the relationships among the variables in the HBALL and HBO models is higher compared to the HBW and HBE models. These models have a wider data range and a higher average daily trip count. As shown in Table 1, the trip range for the HBALL and HBO models is (2–18) and (0–11) with an average of 6.42 and 3.26 trips per day, respectively. The HBW and HBE models, in contrast, have a trip range of (0–5) and (0–7), with an average of 1.56 and 1.59 trips per day, respectively.

The ANFIS approach proves to be superior in developing the HBALL and HBO models compared to the MLR method. The ANFIS approach results in an improvement of 8.33% in the R-squared value for the HBALL model, going from 65.85% to 74.18%. Furthermore, the ANFIS approach leads to a reduction of 13.04%, 19.30%, and 24.39% in RMSE, MAE, and RSS, respectively. The ANFIS approach is able to capture the variation in HBALL trips more accurately than the MLR method. Validation results also show that ANFIS provides more accurate and closer predictions to the actual values compared to MLR. The mean and standard deviation predicted by ANFIS are closer to the actual values than those predicted by MLR. For the HBO model, ANFIS results in a 3.29% improvement in the R-squared value, increasing it from 80.65% to 83.94%. ANFIS also reduces RMSE, MAE, and RSS by 8.61%, 12.26%, and 16.97%, respectively.

For the HBW and HBE models, the performance of both ANFIS and MLR approaches are similar. Neither ANFIS nor MLR shows a significant improvement over the other in terms of R-squared values or error reduction. The R-squared values of the MLR for these models are already high, above 90%, which means that it is capable of capturing most of the variation among the HBW and HBE trips. The difference in performance measures between ANFIS and MLR is minimal, and can be disregarded. Additionally, the validation results indicate that MLR provides more accurate predictions compared to ANFIS, hence it might be the better choice for these models.

6. Conclusions

In this study, the application of the ANFIS in modelling home-based trip generation has been illustrated, using local dataset collected for Salfit City, Palestine. The performance of ANFIS has been evaluated in comparison to the traditional MLR approach. The implementation methodology and underlying principles of the ANFIS architecture have been also discussed. The overall finding of this research is the ability of ANFIS to outperform the traditional MLR approach, especially in modelling systems with complex nonlinear behaviour that encounter interdependent relationships among their components with large dataset and multiple input variables.

The results of the study show that the HBALL model was influenced by the number of employed persons in the household, persons going to school, and those with a driving license. The HBW model was primarily influenced by the number of employed persons, young household members between 17 and 30 years, and the number of cars owned by the household. The HBE model was only influenced by the number of household members who are receiving education. The HBO model was influenced by the household size, the number of individuals with a driving license, and the presence of an independent household type.

The models of HBALL and HBO were developed more accurately with ANFIS, resulting in better performance than the traditional MLR approach. These two models have a larger data range and higher average number of trips per day, and therefore exhibit more complicated behaviour. The performance measures used confirm the higher predictive accuracy of ANFIS compared with MLR. The R-squared value for the HBALL model increased by 8.33% from 65.85% using MLR to 74.18% using ANFIS, while the RMSE and MAE were reduced by 13.04% and 19.30%, respectively. Similarly, the R-squared value for the HBO model increased from 80.65% using MLR to 83.94% using ANFIS, with a decrease in RMSE and MAE by 8.61% and 12.26%, respectively.

Table 3
Optimum ANFIS Configuration Results for Each Model.

Model Type	R-Squared	Epochs No.	Training Error		Testing Error	
			RMSE	MAE	RMSE	MAE
HBALL	74.18%	80	1.4880	1.1203	1.6397	1.3020
HBW	92.74%	20	0.5465	0.3664	0.5884	0.4305
HBE	96.66%	10	0.4020	0.1318	0.6887	0.2560
HBO	83.94%	100	1.4419	1.1252	1.5002	1.0953

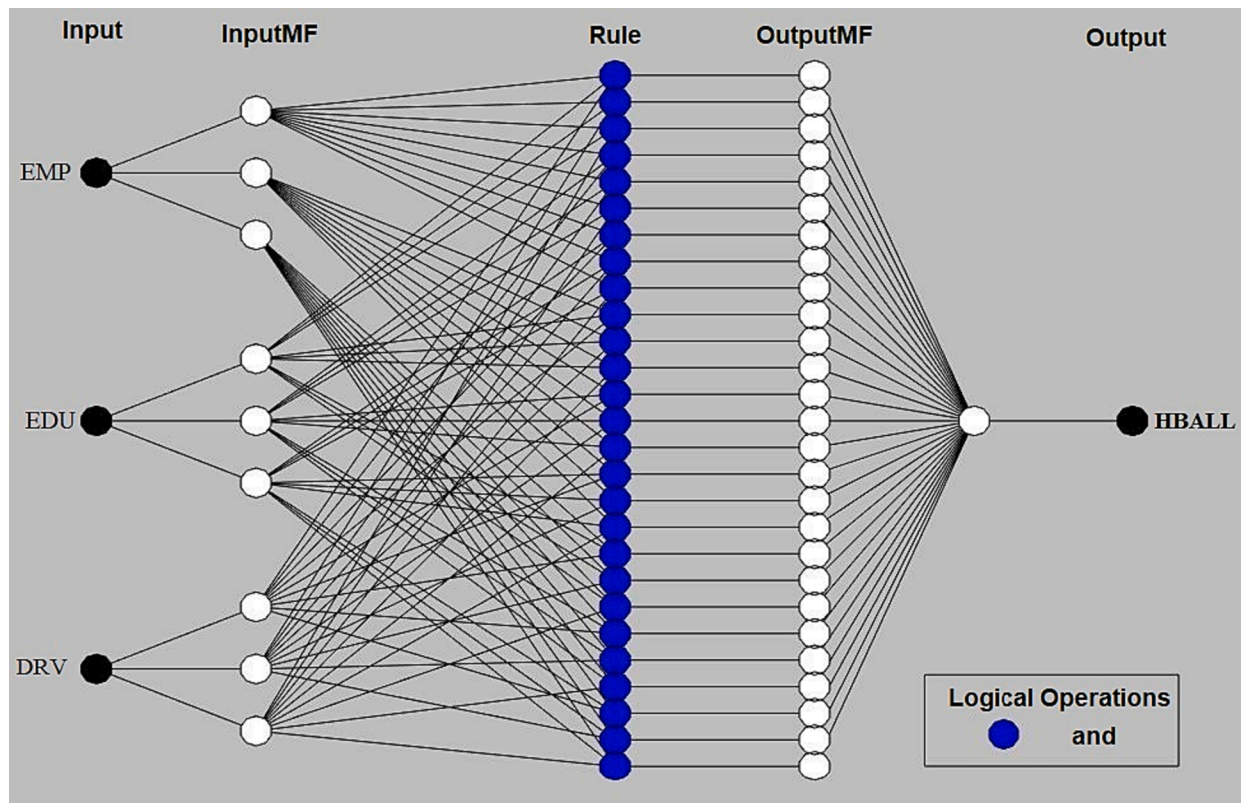


Fig. 3. Equivalent ANFIS Architecture for HBALL.

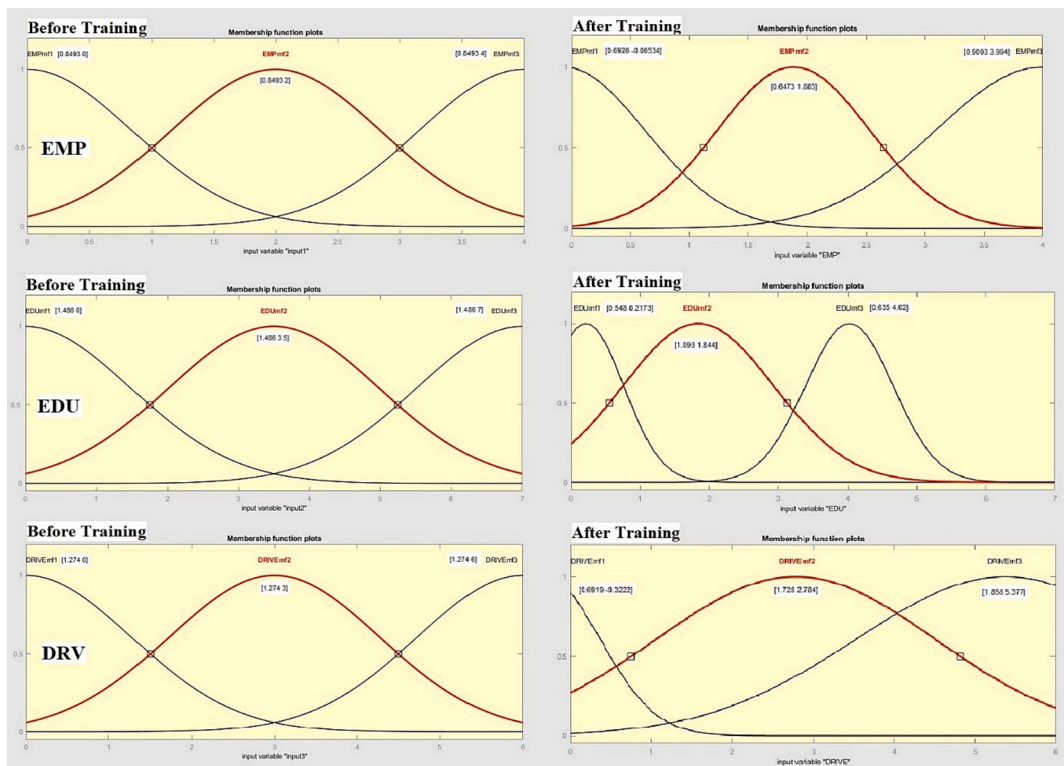


Fig. 4. Initial and Final Gaussian MFs for Each Input Variable in HBALL.

On the other hand, when modelling HBW and HBE, both approaches achieved almost the same accuracy with marginal differences. However, the behaviour of these two models is less complicated as they have a

narrower data range and lower average number of trips per day. The R-squared for both are large enough (i.e., more than 90%) to effectively capture most of the variations among the trips. The differences between

Table 4

Input Gaussian MFs Parameter Before and After Training for Each Model.

Model Type	Variables	Input MFs parameters (σ , μ) before training*						Input MFs parameters (σ , μ) after training*						Input MFs range
		MF1		MF2		MF3		MF1		MF2		MF3		
		σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	
HBALL	EMP	0.849	0	0.849	2	0.849	4	0.692	−0.06	0.647	1.88	0.909	3.99	0–4
	EDU	1.486	0	1.486	3.5	1.486	7	0.548	0.217	1.093	1.844	0.635	4.02	0–7
	DRV	1.274	0	1.274	3	1.274	6	0.692	−0.32	1.728	2.784	1.856	5.377	0–6
HBW	EMP	0.849	0	0.849	2	0.849	4	0.849	0	0.847	2.01	0.855	3.99	0–4
	AGE _B	0.849	0	0.849	2	0.849	4	0.847	0	0.849	1.99	0.855	3.99	0–4
	CAR	0.637	0	0.637	1.5	0.637	3	0.635	0	0.637	1.49	0.637	2.99	0–3
HBE	EDU	1.486	0	1.486	3.5	1.486	7	1.622	0.246	0.473	3.08	1.088	6.832	0–7
HBO	SIZE	1.699	0	1.699	5	1.699	9	1.229	1.764	0.961	6.44	1.557	8.437	0–9
	DRV	1.274	0	1.274	3	1.274	6	0.88	−0.26	1.425	2.65	1.776	5.373	0–6
	HHTYP	0.424	0	0.424	1	–	–	0.338	−0.03	0.246	1.056	–	–	0–1
* Gaussian-type for all input MFs, defined by the standard deviation (σ), and the mean value (μ)														

* Gaussian-type for all input MFs, defined by the standard deviation (σ), and the mean value (μ)**Table 5**

Models Performance Comparison and Validation.

Model Type	Performance Comparison						Models Validation			
	Index	MLR	><	ANFIS	Diff.	% Diff.	Index	Actual	MLR	ANFIS
HBALL	RMSE	1.7112	>	1.4880	-0.2232	-13.04%	RMSE	0	1.9540	1.6397
	MAE	1.3882	>	1.1203	-0.2679	-19.30%	MAE	0	1.4587	1.3020
	RSS	749.60	>	566.786	-182.812	-24.39%	Mean	6.81	6.44	6.64
	R-squared	65.85%	<	74.18%	8.33%	-8.33%	St. Dev.	3.563	2.639	3.026
HBW	RMSE	0.5932	>	0.5465	-0.0467	-7.87%	RMSE	0	0.5788	0.5884
	MAE	0.4163	>	0.3664	-0.0499	-11.99%	MAE	0	0.3582	0.4305
	RSS	90.070	>	76.456	-13.614	-15.11%	Mean	1.45	1.47	1.54
	R-squared	90.36%	<	92.74%	2.38%	-2.38%	St. Dev.	0.932	0.764	0.739
HBE	RMSE	0.4035	>	0.4020	-0.0015	-0.37%	RMSE	0	0.6761	0.6887
	MAE	0.1430	>	0.1318	-0.0112	-7.83%	MAE	0	0.2853	0.2560
	RSS	41.676	>	41.377	-0.3000	-0.72%	Mean	1.58	1.65	1.72
	R-squared	96.63%	<	96.66%	0.03%	-0.03%	St. Dev.	1.69	1.59	1.64
HBO	RMSE	1.5778	>	1.4419	-0.1359	-8.61%	RMSE	0	1.9647	1.5002
	MAE	1.2825	>	1.1252	-0.1572	-12.26%	MAE	0	1.5141	1.0953
	RSS	637.26	>	529.146	-108.117	-16.97%	Mean	3.77	2.78	3.34
	R-squared	80.65%	<	83.94%	3.29%	-3.29%	St. Dev.	2.006	0.974	1.093

the performance measures are marginal and could be neglected. However, utilizing ANFIS as a different approach does not result in any substantial enhancement in modelling trip generation for these two purposes.

The small sample size in the available data source, and the resources constraints for data collection in Palestine, are limiting the ability to generalize the results of this study. However, the comparison and validation processes suggest that ANFIS is a promising modelling approach that can be considered as a strong rival to the traditional MLR approach, especially when dealing with complex relationships that involve multiple socioeconomic variables.

It is recommended to implement the ANFIS approach for mode choice modelling studies and compare its results with traditional techniques, as the behavior of trip-making involves complex decision-making processes with a high degree of uncertainty and ambiguity. ANFIS may prove to be a more effective method for analyzing such systems compared to other techniques already used in transportation planning research. Additionally, it would be interesting to compare ANFIS with other advanced techniques such as deep learning methods and machine learning algorithms, to in-depth evaluate its performance and accuracy.

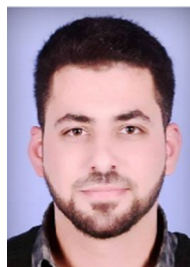
Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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